
THE IMPACT OF VISIBLE HOMELESSNESS ON RETAIL STORE PERFORMANCE IN DOWNTOWN TORONTO: AN ANALYTICAL ASSESSMENT

MBAI 5100: Business Analytics

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1. Definition of the Business Problem

Downtown Toronto experiences visible and persistent levels of homelessness, which causes concerns, and difficulties for local businesses and nearby communities. The appearance of homeless individuals in downtown creates concerns for retailers, business owners, and service-based firms as it might affect customer shopping behavior, which in turn affects their sales results. Multiple stakeholders need information to understand how homelessness may affect retail business operations, and most importantly, their sales.

Therefore, the research question that needs to be investigated is to find the relationship between homelessness in downtown Toronto, local pedestrian traffic, and monthly retail sales; to see if the level of homelessness disrupts the economic activity and pedestrian traffic in downtown Toronto. Most businesses rely on anecdotal articles and reports, stating that higher homelessness may change consumer behaviour, therefore affecting businesses' retail sales; as some customers may not be willing to go to certain stores or areas due to the amount of homeless people in that area. However, without empirical data and research, businesses and organizations risk making costly decisions, as they may base their decisions on assumptions rather than actual evidence.

This analysis helps solve that issue. Multiple stakeholders and business owners depend on actual and accurate insights to see if homelessness meaningfully affects store visits and sales outcomes. This project provides a data-driven assessment, examining the homelessness levels, pedestrian patterns, and retail sales in downtown Toronto. Through statistical analysis, this project will help business owners and stakeholders understand and make decisions based on actual evidence, rather than assumptions or anecdotes.

2. Description of Data:

The City of Toronto Open Data Portal offers three datasets to analyze retail sales, homelessness activities, and pedestrian traffic during 2021 and 2022.

[Toronto Retail Sales Data \(Economic Dashboard\):](#)

- This dataset contains 80 entries, which include two measurement variables.
- The dataset shows Toronto retail sales data in billions of dollars for each month.
- It also shows the current state of Toronto's economic performance.

Daily Shelter Overnight Service Occupancy & Capacity Data (2021 and 2022)

- The dataset contains 100,422 entries which include 32 variables (across 2021 and 2022)
- The dataset monitors shelter usage through daily records, which show locations and service user numbers, and capacity levels.
- It also delivers detailed information about local shelter usage patterns.

Traffic Volumes - Multimodal Intersection Turning Movement Counts

- This dataset contains 6,312 entries, which include 34 different variables.
- The dataset tracks pedestrian and vehicle movements at essential transportation points.
- Researchers can use foot traffic data to evaluate how well retail businesses reach their customer base.

These three datasets help examine the connections between homelessness, pedestrian activity, and retail business success in downtown Toronto.

3. Data Assessment and Preparation:

To investigate whether homelessness in downtown Toronto affects local pedestrian traffic and monthly sales, the project relies on three independent datasets, as stated before. These datasets do provide enough information in addressing the core research question, although the data needs to be cleaned and transformed before analysis.

3.1 Availability of Data:

The data does exist; however, it is spread across multiple sources, and it is reported at different levels (such as monthly or daily). While these datasets are not perfect, they can be transformed into a monthly dataset that can be used for statistical analysis.

Homelessness (Shelter Usage) Data for 2021 and 2022:

- Captures shelter usage counts, which can be used to represent homelessness levels

- Frequency: Daily
- Geographic Detail: Individual shelter sites and postal codes

Pedestrian Foot-Traffic Data:

- Captures downtown pedestrian foot-traffic movement across different intersections
- Frequency: Daily (with some missing counts; counts not available for every month)
- Geographic Detail: Downtown Intersections with latitude/longitude

Retail Sales Data:

- Captures retail sales data per month for the city of Toronto
- Frequency: Monthly, but counts are missing for some months
- Geographic Detail: City of Toronto

3.2 Data Preparation Required to Make the Datasets Analyzable

There were steps involved in transforming the raw data into a usable format for statistical analysis. Firstly, the shelter data was filtered to downtown postal codes, and foot traffic data was filtered based on specific dimensions for latitude and longitude, centering downtown Toronto. Secondly, the daily frequencies for homelessness and pedestrian were converted to monthly, matching the retail sales data. Thirdly, missing pedestrian counts from the dataset were imputed using the median; median from the existing pedestrian counts in the dataset. Lastly, once all the data was cleaned, it was then merged into a single dataset that included homelessness counts, pedestrian foot-traffic counts, and retail sales.

4. Proposed Analysis:

4.1 Overview of the Analytical Approach

As previously stated, the main objective is to investigate the relationship between homelessness in downtown Toronto, local pedestrian traffic, and monthly retail sales, to see if the level of homelessness affects the economic activity and pedestrian traffic in downtown Toronto. To address this, the analysis follows the structured process, which includes data preparation and cleaning, performing linear regression, and cross-validation, as this will help

illustrate the relationships, patterns, and trends that exist between these three variables (homelessness, pedestrian traffic, and retail sales).

4.2 Data Preparation and Wrangling

4.2.1 Filtering by Geographic Relevance:

Since the research question is focused on downtown Toronto, the analysis filtered out the shelter records based on postal code, which included M5, M4X, and M4Y. This allowed the analysis to focus on specifically downtown Toronto, and not anywhere else. Similarly, the traffic dataset was filtered using latitude and longitude boundaries corresponding to the same central area. This ensured that all variables were focused on the same area.

4.2.2 Converting Daily Data Into Monthly Metrics:

Both the shelter and traffic datasets are reported at a daily level, as the retail dataset is available monthly. For the analysis, the shelter and traffic variables had to be converted to the monthly level, as this smoothens out short-term fluctuations that may appear, and it matches the retail dataset. This was done by using the mean daily count for each month.

4.2.3 Addressing Missing Data Through Imputation:

When looking at the traffic dataset, it did not contain pedestrian counts for every month in 2021 to 2022. As a result, when converting the traffic counts to the monthly frequency from daily, some months had missing values. Instead of dropping the months, that dataset was reindexed to include every month in the analysis period. The missing pedestrian counts were then imputed by using the median pedestrian count (observed from the available months); as by using the median can help avoid any outliers.

4.2.4 Merging the Datasets:

After cleaning, converting, and aligning the datasets, they were then all merged into a single table indexed by the month. This merged dataset will help with future analysis/modelling.

4.3 Descriptive Statistics and Visualization

Summary statistics were computed to help establish an understanding for all the variables, as this included the mean, median, range, etc. In addition to the statistics computed, three line charts were generated for each variable, as each line graph illustrates a clear visual representation of how each measure is evolved over time.

4.4 Multiple Linear Regression

To see if there is a relationship whether homelessness affects economic activity, a multiple linear regression model was performed, by using monthly retail sales as the outcome, and homelessness and foot traffic as predictors. A multiple linear regression was used because it can quantify and highlight whether if higher homelessness in downtown tends to experience higher or lower retail sales, while holding foot traffic constant, so that the effect of homelessness is isolated. By doing so, it indicates the approximate magnitude of the relationship between the variables.

4.5 Cross-Validation for Model Reliability

A year-based cross-validation was used to see if the regression model performs beyond the sample data in which it was trained. Data from 2021 served as the training set, and 2022 as the testing set. The model was fit using the training set (2021 data), and then evaluated based on its predictions for 2022. This was essential, as it demonstrates whether the model is predictive or not.

5. Expected Results:

5.1 Observed Monthly Patterns

5.1.1 Homelessness Trends

The monthly homelessness time series showed moderate variability; however, there was no clear long-term upward or downward trend. Shelter usage levels did fluctuate month to month, but over the course of the two-year window, it remained broadly stable. This suggests that homelessness stayed relatively stable and did not change in a major or long-lasting way.

5.1.2 Foot Traffic Trends

Foot traffic, however, changed a lot month to month, as in some months it would rise quickly, and in other months it would decrease. These rapid changes from month to month do not match the pattern shown of the homelessness levels; when there is an increase in foot traffic, it does not necessarily mean homelessness spiked, and vice versa. This implies that foot traffic is influenced by other major factors, which could include weather, events, tourism, etc.

5.1.3 Retail Sales Trends

Based on the trends displayed, retail sales steadily increased over the time period, as it also shows predictable seasonal patterns, such as sales increasing during holidays, etc. However, even though retail sales were rising and fluctuating, these changes did not match or were not in sync with homelessness levels. This suggests that homelessness does not appear to influence retail sales, as there were other major external factors involved.

5.2 Regression Results: Limited Predictive Relationship

Overall, the regression results indicated that retail sales are not meaningfully predicted by the level of homelessness or foot traffic. The regression model illustrated a weak overall model fit, as the R-squared was 0.198; this means the model explains/predicts 20% of the variation in monthly retails. The other 80% are from other external factors that aren't included in the model. Also, homelessness and pedestrian count have non-significant coefficients (high p-values), as both are above 0.05 (the significance threshold). This means that the model cannot explicitly predict that increasing or decreasing homelessness will affect retail sales, as there is not a solid relationship between foot traffic and retail sales. Lastly, the confidence interval for homelessness and foot traffic both include positive and negative numbers, and this means that the model is not sure whether homelessness or foot traffic increases or decreases retail sales.

5.3 Cross-Validation Results:

The cross-validation results indicated that the model trained on 2021 does not work enough for predicting 2022. In terms of the training data, R-squared was 0.521, which means

that the model was able to explain/predict around 52% of the changes in retail sales in 2021. Train RMSE was 486 million dollars, which is large, but in terms of provincial retail numbers, it is not extremely high (it was moderate). However, in terms of testing data, the model performed poorly in 2022. Validation RMSE was 1.83 billion dollars, around 4 times more than the train RMSE. This indicates that the patterns and relationships seen in 2021 did not hold in 2022, and therefore homelessness and foot traffic do not have a solid year-to-year effect on retail sales.

5.4 Homelessness and Foot Traffic: No Significant Relationship

Based on the linear regression and cross-validation, there is no significant relationship linking between homelessness levels and foot traffic. Having a near-zero R-squared and non-significant coefficients illustrates that changes and variation in shelter usage do not relate to consistent increases or decreases in foot traffic. Foot traffic downtown displayed in the model, is driven by external factors not present in the model, such as city events, tourism, etc.

6. Overall Conclusion:

After bringing all the results together, the analysis concludes that homelessness levels in downtown Toronto do not illustrate a significant relationship with either retail sales or foot-traffic patterns within the 2021 to 2022 time period. Foot traffic patterns occur independently from homelessness levels, and retail sales in downtown Toronto are dictated by external economic factors rather than the change of homelessness and foot traffic levels. Therefore, the predictive models that were used in the analysis did not generalize well.

Despite this setback, there are several future improvements that could be done to offer better insights and understanding. Firstly, having a longer time period would definitely help illustrate broader trends and relationships, as it would improve the statistics and accuracy of the predictive models. Secondly, by using more granular retail data, such as using a store-level sales rather than the entire retail sales of the city of downtown Toronto, would better reflect the local impacts of homelessness. Lastly, if possible, adding extra external variables, such as city events, or economic indicators, would help improve the accuracy of pedestrian activity and retail

sales. This can be implemented by finding the correct dataset for the variable/factor that is being added; it would then be cleaned, converted to the same monthly frequency, and then merged into the existing dataset. This would then account for the other external factors that influence downtown activity.

Understanding the relationship between homelessness, pedestrian traffic, and retail activity is important for decision-making for business owners and stakeholders, as the insights from this analysis can help them make correct decisions, and address social challenges while supporting and improving Toronto's economy.

Appendix:

Figure A1: Monthly Downtown Homelessness (2021-2022)

This figure illustrates the average daily usage of shelters per month for downtown Toronto. Shooter usage spiked during the end of 2021 and the end of 2022.

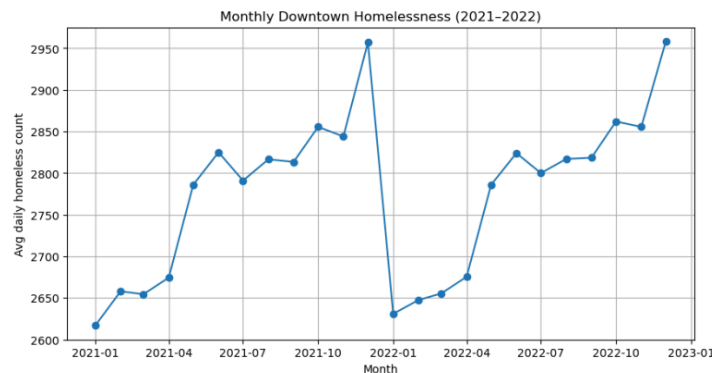


Figure A2: Monthly Downtown Foot Traffic (2021-2022)

This figure illustrates the pedestrian traffic, as pedestrian count has sharply increased and decreased throughout the months. These fluctuations do not align with homelessness levels, suggesting that foot traffic is driven by external factors.

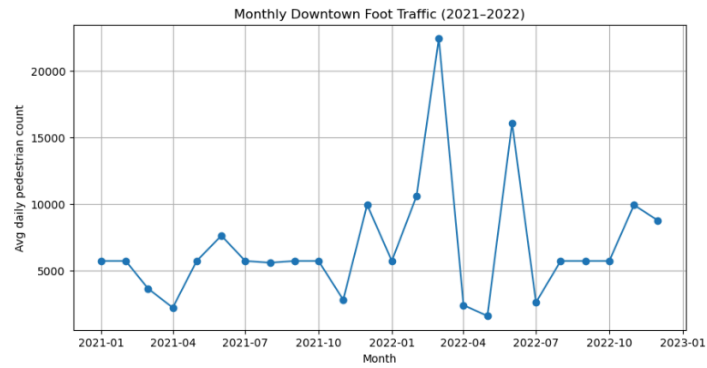


Figure A3: Monthly Downtown Foot Traffic (2021-2022)

This figure illustrates retail sales, as it displays seasonal peaks, around the holiday season, like Christmas. There is a gradual upward recovery trend, following the pandemic. However, this pattern doesn't match with shelter usage trends, suggesting that homelessness is not a significant predictor of sales.



Figure A4: Regression Plot

This figure illustrates the regression model, as it examines whether downtown homelessness levels and pedestrian traffic impacted and predicted retail sales in Toronto. The model generated an R-squared of 0.198, and an adjusted R-squared of 0.122, suggesting that the predictors explain a small portion of variation in retail sales. Both homelessness ($p = 0.158$) and pedestrian counts ($p = 0.102$) showed no significant effects, as their confidence intervals include zero. This means that there is very little to no solid evidence that changes in homelessness affect monthly sales.

Rows used for regression: 24

OLS Regression Results						
Dep. Variable:	retail_dollars		R-squared:	0.198		
Model:	OLS		Adj. R-squared:	0.122		
Method:	Least Squares		F-statistic:	2.591		
Date:	Wed, 03 Dec 2025		Prob (F-statistic):	0.0987		
Time:	00:37:48		Log-Likelihood:	-526.70		
No. Observations:	24		AIC:	1059.		
Df Residuals:	21		BIC:	1063.		
Df Model:	2					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
Intercept	2.523e+09	5.1e+09	0.495	0.626	-8.08e+09	1.31e+10
homeless_count	2.689e+06	1.84e+06	1.465	0.158	-1.13e+06	6.51e+06
pedestrian_count	6.769e+04	3.96e+04	1.711	0.102	-1.46e+04	1.5e+05
Omnibus:	0.867	Durbin-Watson:	0.585			
Prob(Omnibus):	0.648	Jarque-Bera (JB):	0.876			
Skew:	-0.325	Prob(JB):	0.645			
Kurtosis:	2.327	Cond. No.	2.42e+05			

The formula used:

$$\text{retail_dollars} = \beta_0 + \beta_1(\text{homeless_count}) + \beta_2(\text{pedestrian_count}) + \epsilon$$

Figure A5: Cross Validation

Cross-validation was done afterwards, to assess whether the regression model generalizes across the years. The model was trained using 2021 data and validated using 2022 data. The model produced an R-squared of 0.521 and an adjusted R-squared of 0.414, suggesting a moderate fit on the training period. However, despite the moderate fit, the model failed to generalize the following year, as the training RMSE was \$486 million but the validation RMSE was \$1.83 billion, indicating major prediction error. This means, in result, that the model is not solid nor reliable due to the huge discrepancy.

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Train rows (2021): 12
Validation rows (2022): 12

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                        OLS Regression Results
=====
Dep. Variable:          retail_dollars    R-squared:                0.521
Model:                  OLS              Adj. R-squared:           0.414
Method:                 Least Squares    F-statistic:             4.889
Date:                   Wed, 03 Dec 2025  Prob (F-statistic):      0.0365
Time:                   00:37:48         Log-Likelihood:          -257.06
No. Observations:       12              AIC:                    520.1
Df Residuals:           9               BIC:                    521.6
Df Model:               2
Covariance Type:        nonrobust
=====
                        coef      std err      t      P>|t|      [0.025      0.975]
-----
Intercept              -6.049e+09   5.25e+09   -1.152    0.279   -1.79e+10   5.83e+09
homeless_count         5.804e+06   1.99e+06    2.912    0.017    1.29e+06   1.03e+07
pedestrian_count      -6.407e+04   9.9e+04   -0.647    0.534   -2.88e+05   1.6e+05
=====
Omnibus:                6.616    Durbin-Watson:           1.277
Prob(Omnibus):          0.037    Jarque-Bera (JB):        3.131
Skew:                   -1.192    Prob(JB):                0.209
Kurtosis:               3.763    Cond. No.                2.09e+05
=====

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 2.09e+05. This might indicate that there are
strong multicollinearity or other numerical problems.

Model Performance:
Train RMSE (2021): 486,416,922.43
Validation RMSE (2022): 1,830,438,439.84
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The datasets and code are found in this Google Drive link:

https://drive.google.com/drive/folders/1mDZS8qRvlfP1FEiMNFF0GalBzwWfwZ_T?usp=drive_link